

PROGRAM:

```
#include <stdio.h>

void insertionSort(int arr[], int n) {
    int i, key, j;
    for (i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
}

void printArray(int arr[], int n) {
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    printf("\n");
}

int main() {
    int choice, n, i;
    printf("\n-----INSERTION SORT      \n ");
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d elements are :\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);
}
```

```

do {

    printf("\nMenu:\n");

    printf("1. Sort using Insertion Sort\n");

    printf("2. Print the sorted array\n");

    printf("3. Exit\n");

    printf("Enter your choice: ");

    scanf("%d", &choice); switch (choice) {

    case 1:

        insertionSort(arr, n);

        printf("Array sorted using Insertion Sort.\n");

        break;

    case 2:

        printf("Sorted array: ");

        printArray(arr, n); break;

    case 3:

        printf("Exiting the program.\n");

        break;

    default:

        printf("Invalid choice! Please enter a valid menu option.\n");

    }

} while (choice != 3);

return 0;

}

```

OUTPUT:

-----INSERTION SORT

Enter the number of elements: 4

Enter 4 elements are :

1

56

23

78

Menu:

1. Sort using Insertion Sort

2. Print the sorted array

3. Exit

Enter your choice: 1

Array sorted using Insertion Sort.

Menu:

1. Sort using Insertion Sort

2. Print the sorted array

3. Exit